

B.Tech. Degree IV Semester Regular/Supplementary Examination June 2024

CE/CS/EC/EE/IT/ME/SE

19-200-0401 COMPLEX VARIABLES AND PARTIAL DIFFERENTIAL EQUATIONS

(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Transform a region to another using conformal mapping.
 CO2: Evaluate real integrals using Residue theorem.
 CO3: Form and solve partial differential Equations.
 CO4: Determine solution of partial differential equation for vibrating string and heat conduction.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate,
 L6 – Create

PO – Programme Outcome

PART A

(Answer ALL questions)

(8 × 3 = 24) Marks BL CO PO

- I. (a) Show that $u = x^2 - y^2 - 2xy - 2x + 3y$ is harmonic. 3 L3 1 2
 (b) Find the image of $|z - 1| = 1$ under the mapping $w = 1/z$. 3 L2 1 2
 (c) Using Cauchy's integral formula, 3 L3 2 3
 evaluate $\int_c \frac{e^{-z}}{z^2} dz$.
 (d) Expand $\frac{1}{z^2 - 3z + 2}$ in the region $1 < |z| < 2$. 3 L3 2 3
 (e) Solve $p^2 = p + q$. 3 L2 3 2
 (f) Solve $z = p^2 + q^2$. 3 L2 3 2
 (g) Using the method of separation of variables, solve 3 L2 4 2
 $\frac{\partial u}{\partial x} = 2 \frac{\partial u}{\partial t} + u$; given $u(x, 0) = 6e^{-3x}$.
 (h) Find the solution of the one dimensional wave equation. 3 L1 4 1

PART B

(4 × 12 = 48)

- II. (a) Show that $f(z) = \sqrt{XY}$ is not analytic at the origin, although C-R 6 L3 1 3
 equations are satisfied at that point. 6 L3 1 3
 (b) Find the analytic function $f(z)$ whose real part is
 $e^{2x}(x \cos 2y - y \sin 2y)$.

OR

- III. (a) Determine the analytic function $w = u + iv$ 6 L3 1 3
 where $v = \log(x^2 + y^2) + x - 2y$. 6 L2 1 2
 (b) Find the bilinear transformation which maps the points $z = 2, i, -2$ in
 to points $w = 1, i, -1$.

(P.T.O.)

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		Marks	BL	CO	PO
IV.	(a) Verify Cauchy's theorem for the integral of z^3 taken over the boundary of the rectangle with vertices $-1, 1, 1+i, -1+i$.	6	L2	2	2
	(b) Expand $\frac{z-1}{z^2}$ in powers of $z-1$ and determine the region of convergence.	6	L3	2	3
OR					
V.	(a) Using Cauchy's Residue Theorem, evaluate $\int_c \frac{z}{(z-1)(z-2)^2} dz$ where c is $ z-2 = \frac{1}{2}$.	6	L3	2	3
	(b) Evaluate $\int_0^{2\pi} \frac{d\theta}{2 + \cos \theta}$.	6	L2	2	2
VI.	(a) Find the partial differential equation of all spheres whose centres lie on z -axis.	6	L3	3	2
	(b) Solve $r - 4s + 4t = \sin(3x + 4y)$.	6	L3	3	3
OR					
VII.	(a) Find the partial differential equation by eliminating the arbitrary function from $f(x+y+z, x^2+y^2+z^2) = 0$.	6	L2	3	2
	(b) Solve $x(y-z)p + y(z-x)q = z(x-y)$.	6	L2	3	3
VIII.	(a) Solve $Uxx - Uy = 0$ by the method of separation of variables.	6	L1	4	1
	(b) A string is stretched and fastened to two points 1 apart. Motion is started by displacing the string in the form $y = a \sin\left(\frac{\pi x}{l}\right)$ from which it is released at time $t = 0$. Show that the displacement of any point at a distance x from one end at time t is given by $y(x, t) = a \sin\left(\frac{\pi x}{l}\right) \cos\left(\frac{\pi ct}{l}\right)$.	6	L4	4	4
OR					
IX.	(a) Solve the heat equation.	6	L1	4	1
	(b) Solve Laplace's equation in two dimensions.	6	L3	4	3

Blooms's Taxonomy Level's

L1 - 15%, L2 - 45%, L3 - 25%, L5 - 15%.
